FIME
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA

Arquitectura de Robots

Robots as Mechanical Devices

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Section 1.1

Classification of Robotic Manipulators

Classification of Robotic Manipulators

Robot manipulators can be classiffed by several criteria:

Power source, meaning the way in which the joints are actuated;

Geometry, or kinematic structure; their **method of control**; and their intended **application area**.

For example, a hydraulic robot would not be suitable for food handling or clean room applications, whereas a SCARA robot would not be suitable for automobile spray-painting.

Section 1.2

Power Source

Power Source

Most robots are either electrically, hydraulically, or pneumatically powered.

Hydraulic actuators are unrivaled in their speed of response and torque producing capability.

Robots driven by DC or AC motors are increasingly popular since they are cheaper, cleaner, and quieter.

Method of Control

Robots are categorized based on their control mechanisms into **servo-controlled** and **non-servo-controlled** systems.

Nonservo robots are fundamentally **open-loop** systems, characterized by their motion constraints imposed by predefined mechanical limits. These robots are predominantly utilized for the transfer of materials.

Servo robots use **closed-loop** computer control to determine their motion and are thus capable of being truly multifunctional, reprogrammable devices.

Servo controlled robots

Servo controlled robots are further classified according to the method that the controller uses to guide the end effector.

The simplest type of robot in this class is the **point-to-point** robot. A point-to-point robot can be taught a discrete set of points, but there is no control of the path of the end effector in between taught points.

With **continuous path robots**, on the other hand, the entire path of the end effector can be controlled. In addition, the velocity and/or acceleration of the end effector can often be controlled.

Geometry

The majority of contemporary industrial manipulators are characterized by possessing six or fewer degrees of freedom (DOF).

These manipulators are usually classified kinematically on the basis of the first three joints of the arm, with the wrist being described separately.

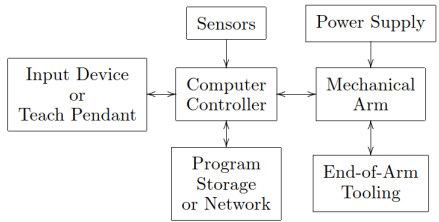
The majority of these manipulators can be classified into one of five geometric configurations: articulated (RRR), spherical (RRP), SCARA (RRP), cylindrical (RPP), or Cartesian (PPP).

Section 1.3

Robotic Systems

The mechanical arm is just one component in an overall robotic system, illustrated in Figure 1, which consists of the arm, external **power source**, end-of-arm **tooling**, **external and internal sensors**, **computer interface**, and **control computer**.

Figura 1: *The integration of a mechanical arm.*



Accuracy and Repeatability

The **accuracy** of a manipulator is a measure of how close the manipulator can come to a given point within its workspace.

Repeatability is a measure of how close a manipulator can return to a previously taught point.

Accuracy and Repeatability affections

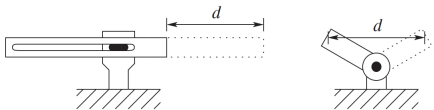
Accuracy is affected therefore by computational errors, machining accuracy in the construction of the manipulator, flexibility effects such as the bending of the links under gravitational and other loads.

Repeatability therefore is affected primarily by the controller resolution. **Controller resolution** means the smallest increment of motion that the controller can sense. The resolution is computed as the total distance traveled divided by $2n$, where n is the number of bits of encoder accuracy.

Linear vs. rotational link

Linear vs. rotational link motion, showing that a smaller revolute joint can cover the same distance d as a larger prismatic joint. The tip of a prismatic link can cover a distance equal to the length of the link. The tip of a rotational link of length d, by contrast, can cover a distance of $2d$ by rotating 180 degrees.

Figura 2: *Linear vs. rotational link motion.*



Wrists and End Effectors

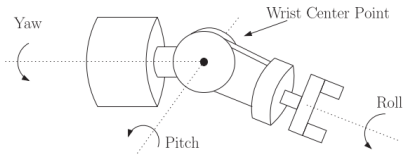
1 Robots as Mechanical Devices

The joints in the kinematic chain between the arm and end effector are referred to as the wrist.

The wrist joints are nearly always all revolute. It is increasingly common to design manipulators with spherical wrists.

Wrists whose three joint axes intersect at a common point, known as the wrist center point.

Figura 3: *The spherical wrist.*



Wrists and End Effectors

Typically, the manipulator will possess three DOF for position, which are produced by three or more joints in the arm.

The arm and wrist assemblies of a robot are used primarily for positioning the hand, end **effector**, and any tool it may carry.

The simplest type of end effector is a **gripper**, which is usually capable of only two actions, **opening** and **closing**.

Figura 4: *The spherical wrist axes.*

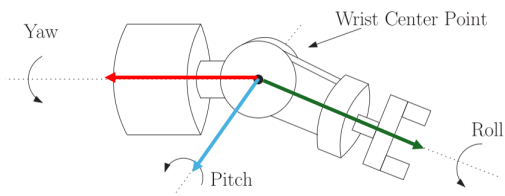




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2 Common Kinematic Arrangements

- Robots as Mechanical Devices

- Common Kinematic Arrangements

Section 2.1

Articulated Manipulator (RRR)

Articulated Manipulator (RRR)

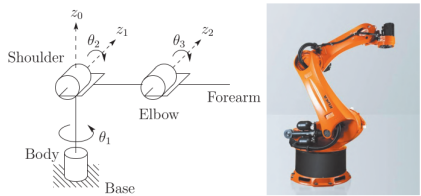
2 Common Kinematic Arrangements

The articulated manipulator is also called a **revolute**, **elbow**, or **anthropomorphic** manipulator.

In the anthropomorphic design, the three links are designated as the body, upper arm, and forearm, respectively.

The joint axes are designated as the waist (z_0), shoulder (z_1), and elbow (z_2).

Figura 5: *Symbolic representation of an RRR manipulator.*



Section 2.2

Spherical Manipulator (RRP)

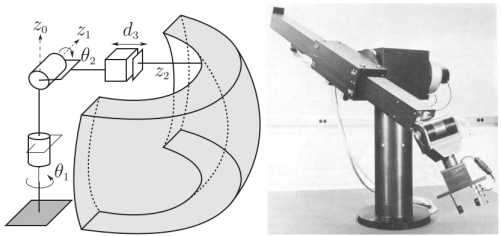
Spherical Manipulator (RRP)

2 Common Kinematic Arrangements

Upon replacing the third joint, or elbow joint, in the revolute manipulator with a prismatic joint, the spherical manipulator is obtained.

The term **spherical manipulator** derives from the fact that the joint coordinates coincide with the spherical coordinates of the end effector relative to a coordinate frame located at the shoulder joint

Figura 6: *Schematic representation of an RRP manipulator.*



Section 2.3

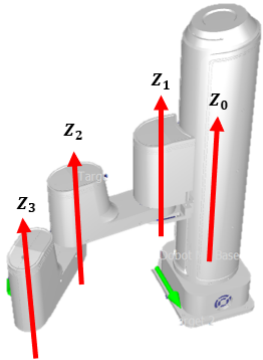
SCARA Manipulator (RRP)

SCARA Manipulator (RRP)

The SCARA arm (for Selective Compliant Articulated Robot for Assembly) is a popular manipulator, which, as its name suggests, is tailored for pick-and-place and assembly operations.

The SCARA has z_0, z_1, and z_2 mutually parallel.

Figura 7: *SCARA robot (RRPR).*



Section 2.4

Cylindrical Manipulator (RPP)

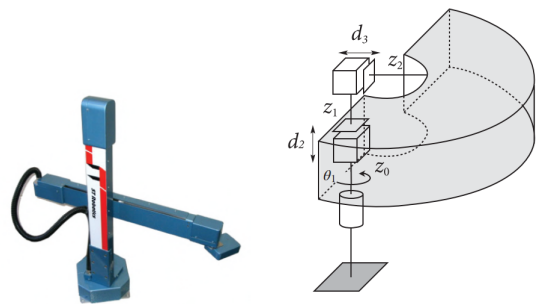
Cylindrical Manipulator (RPP)

The cylindrical manipulator is shown in the figure 8.

The first joint is revolute and produces a rotation about the base, while the second and third joints are prismatic.

The joint variables are the cylindrical coordinates of the end effector with respect to the base.

Figura 8: *ST Robotics R19 cylindrical robot (RPP).*



Section 2.5

Cartesian Manipulator (PPP)

Cartesian Manipulator (PPP)

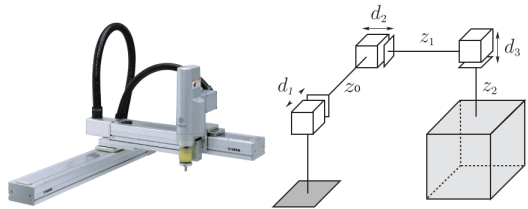
2 Common Kinematic Arrangements

The joint variables of the Cartesian manipulator are the Cartesian coordinates of the end effector with respect to the base.

The kinematic description of this manipulator is the simplest of all manipulators.

Cartesian manipulators are useful for tabletop assembly applications and, as gantry robots, for the transfer of material or cargo.

Figura 9: *The Yamaha YK-XC Cartesian robot (PPP).*



Section 2.6

Parallel Manipulator

Parallel Manipulator

2 Common Kinematic Arrangements

A parallel manipulator is one in which some subset of the links forma closed chain.

A parallel manipulator has two or more kinematic chains connecting the base to the end effector.

The kinematic characterization of parallel robots significantly differs from that of serial link robots, necessitating distinct analytical approaches.

Figura 10: *The ABB IRB360 parallel robot.*



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Q&A

Thank you for your attention

Rodriguez, J. Fundamentos De Robotica Antonio Barrientos.
Craig, J. J. (2006). Introduction to robotics. Pearson Educacion..
Spong, M. W., Hutchinson, S., & Vidyasagar, M. (2020). Robot modeling and control. John Wiley & Sons.